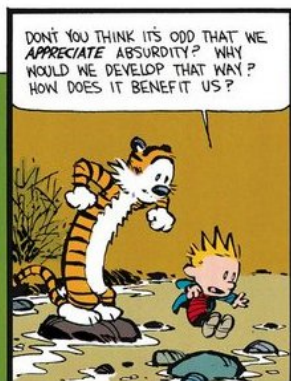
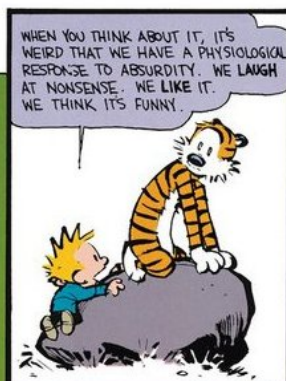
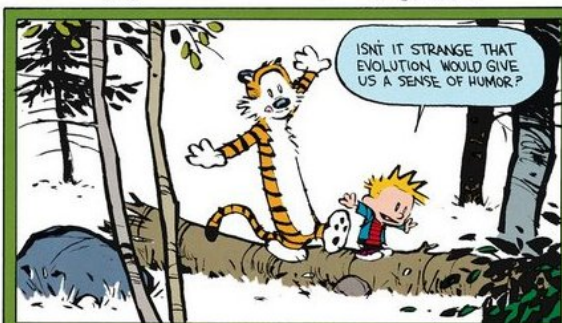
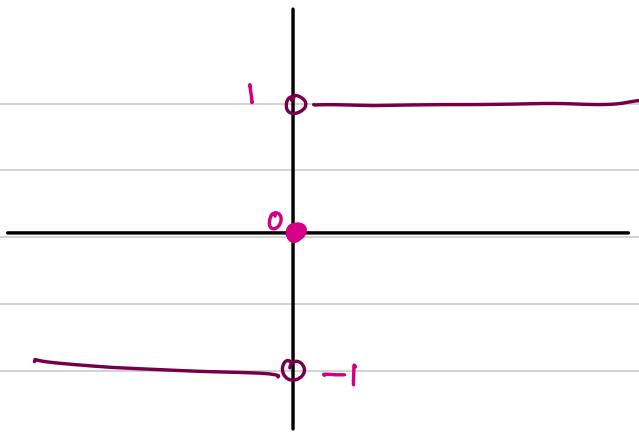


# calvin and Hobbess



$$\lim_{n \rightarrow 8^+} ; n = 8.0000001$$

$$\lim_{n \rightarrow 2^-} ; n = 1.9999999$$



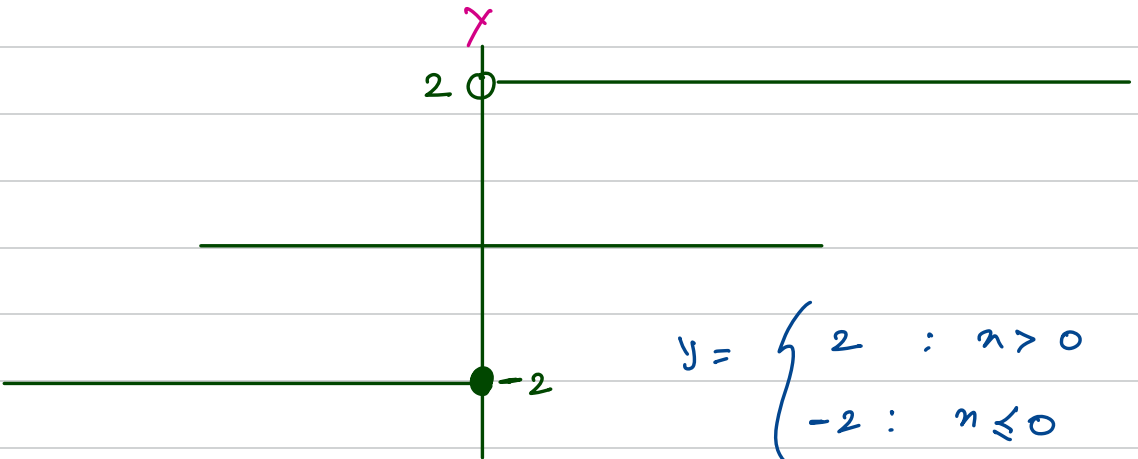
$$y = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} y = -1$$

$$(x = -0.0000001)$$

$$\lim_{x \rightarrow 0^+} y = 1$$

$$(x = 0.0000001)$$

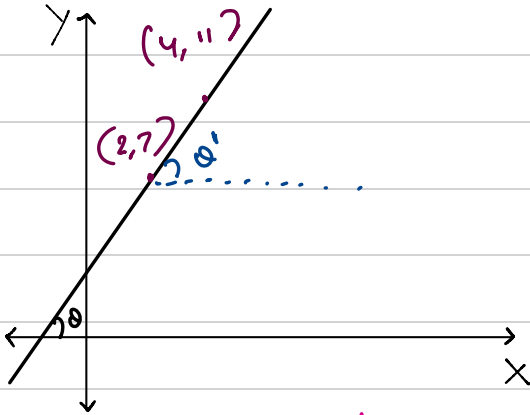


$$y = \begin{cases} 2 & : x > 0 \\ -2 & : x \leq 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} y = -2$$

$$; \quad \lim_{x \rightarrow 0^+} y = 2$$

# Refresher - Slope!? (Again???)



$$y = 2x + 3$$

if I move one unit  
on  $x$ ,  $y$  will  
change by 2 units

$$\theta = \theta'$$

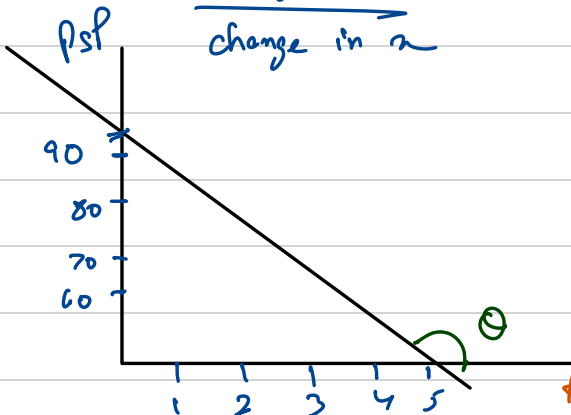
$$\begin{matrix} x_2 & y_2 & x_1 & y_1 \\ (4, 11) & & (2, 7) & \end{matrix} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{11 - 7}{4 - 2} = \frac{4}{2} = 2$$

$$m = \frac{dy}{dx}$$

$d \rightarrow$  delta  $\rightarrow$  change

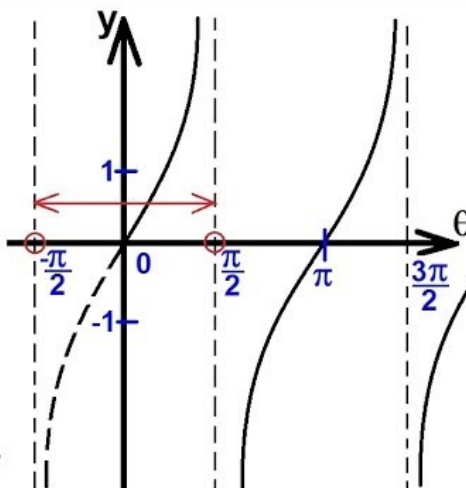
$$= \frac{\text{change in } y}{\text{change in } x}$$

Rate at which  $y$  increases  
with  $x$



# of hours you spend on  
Ph...

| $\theta$<br>(degree) | $\theta$<br>(radian) | $\tan\theta$          |          |
|----------------------|----------------------|-----------------------|----------|
| 0                    | 0                    | 0                     | 0        |
| 30                   | $\frac{\pi}{6}$      | $\frac{1}{\sqrt{3}}$  | 0.577    |
| 45                   | $\frac{\pi}{4}$      | 1                     | 1        |
| 60                   | $\frac{\pi}{3}$      | $\sqrt{3}$            | 1.732    |
| 90                   | $\frac{\pi}{2}$      | $\infty$              | $\infty$ |
| 120                  | $\frac{2\pi}{3}$     | $-\sqrt{3}$           | -1.732   |
| 135                  | $\frac{3\pi}{4}$     | -1                    | -1       |
| 150                  | $\frac{5\pi}{6}$     | $-\frac{1}{\sqrt{3}}$ | -0.577   |



## 1. Core Idea

- **Classification**

Predicts a **discrete label** (category).

Example: spam vs. not spam, digit 0–9, disease vs. no disease.

- **Regression**

Predicts a **continuous value**.

Example: house price, temperature, stock return.

The *same* input space, different output space.

## 2. Type of Output

- **Classification**

- Outputs: class labels, or class probabilities.
- Target  $y$  is from a **finite set**:  $\{0, 1\}$ ,  $\{cat, dog, horse\}$ , etc.
- Often uses an **argmax** over predicted probabilities to pick a label.

- **Regression**

- Outputs: real numbers (possibly bounded in practice).
- Target  $y \in \mathbb{R}$  (or  $\mathbb{R}^k$  for multivariate).
- No argmax; the numeric output itself is the prediction.

# Mean Squared Error

$$\frac{1}{N} \sum_{i=1}^N (\bar{y} - \hat{y})^2$$

| Years | Salary |
|-------|--------|
| 1     | 30     |
| 2     | 40     |
| 3     | 50     |
| 4     | 60     |

$$y = mx + c$$

$$y = w \times x + 20$$

$$y = 10 \times x + 20$$

$$\frac{1}{N} \sum_{i=1}^N (\bar{y} - \hat{y})^2$$

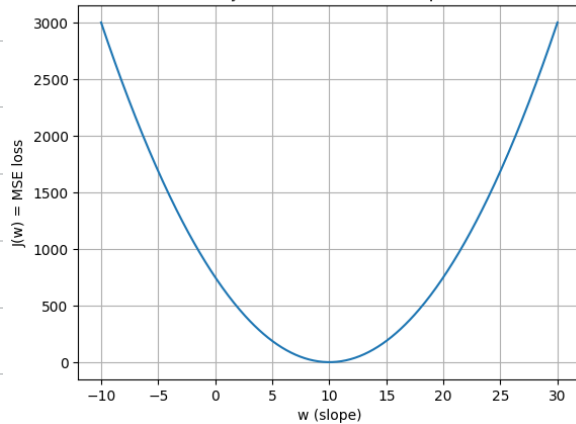
$$L(w=30)$$

| Years | Salary | $\hat{y}(w=0)$ | $L(w=0)$   | $\hat{y}(w=30)$ | $L(w=30)$   |
|-------|--------|----------------|------------|-----------------|-------------|
| 1     | 30     | 20             | 100        | 50              | 400         |
| 2     | 40     | 20             | 400        | 80              | 1600        |
| 3     | 50     | 20             | 900        | 110             | 3600        |
| 4     | 60     | 20             | 1600       | 140             | 6400        |
|       |        |                | <u>750</u> |                 | <u>3000</u> |

$$= 1/4 * (100 + 400 + 900 + 1600)$$

$$= 1/4 * (400 + 1600 + 3600 + 6400)$$

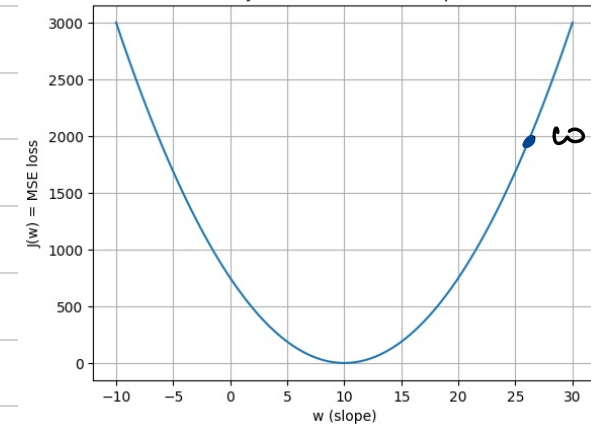
Loss  $J(w)$  as a function of slope  $w$



1. We made 400 models/equations.
2. We did predictions using each of the 400 equations.
3. Compute MSE-Loss for each of the 400 equations.
4. Plot

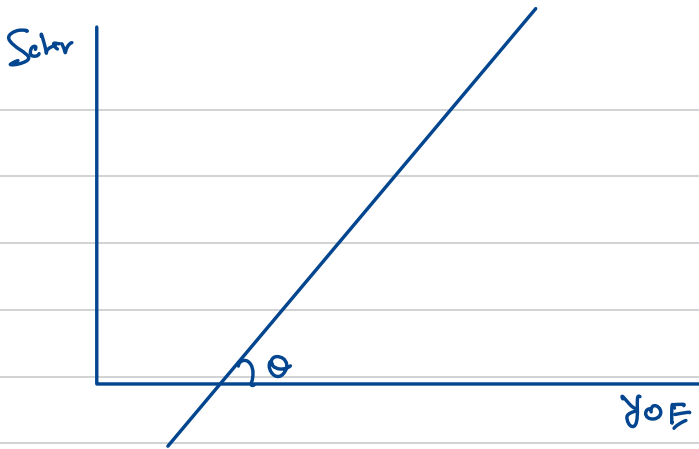
Even if, for a real world, we can't get a loss function curve like this, it still exists.

Loss  $J(w)$  as a function of slope  $w$



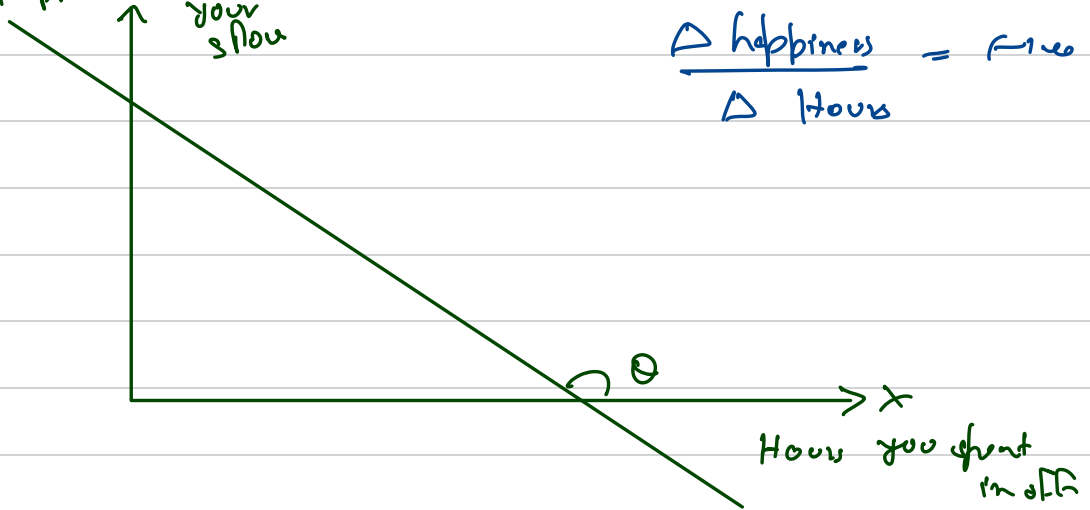
Rate of change of loss wrt  $w$   
 if I increase  $w$ ,  
 loss increases

Slope for  $w$  at any point  $\rightarrow$  rate of increment of loss wrt  $w$ ; then we can go in direction in which loss-decreases,



$$\frac{\Delta \text{Schr}}{\Delta yOE} = \text{rise}$$

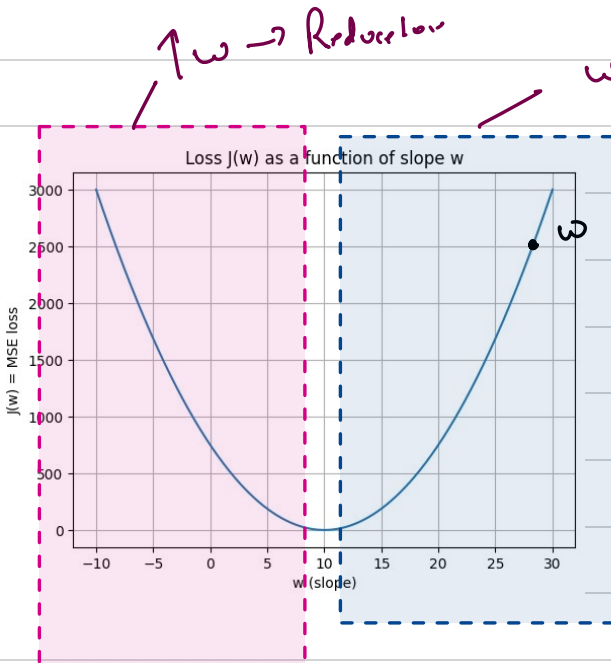
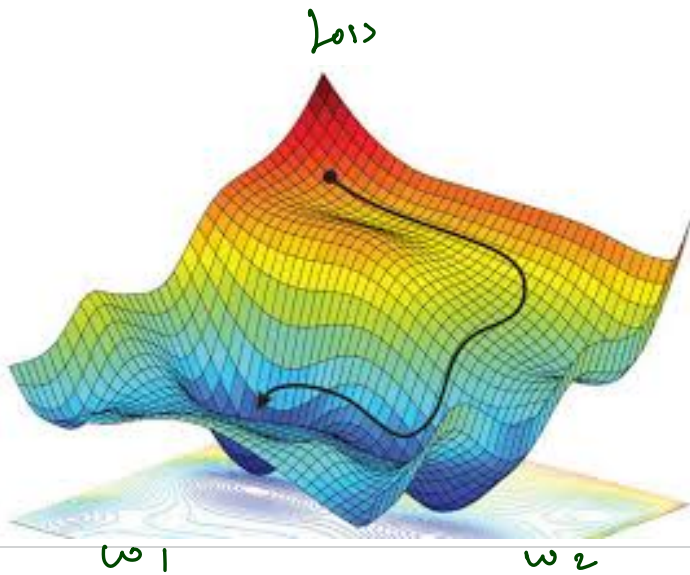
Happiness meter of  
your  
flow



$$\frac{\Delta \text{happiness}}{\Delta \text{Hours}} = \text{rise}$$

Hours you spent  
in off

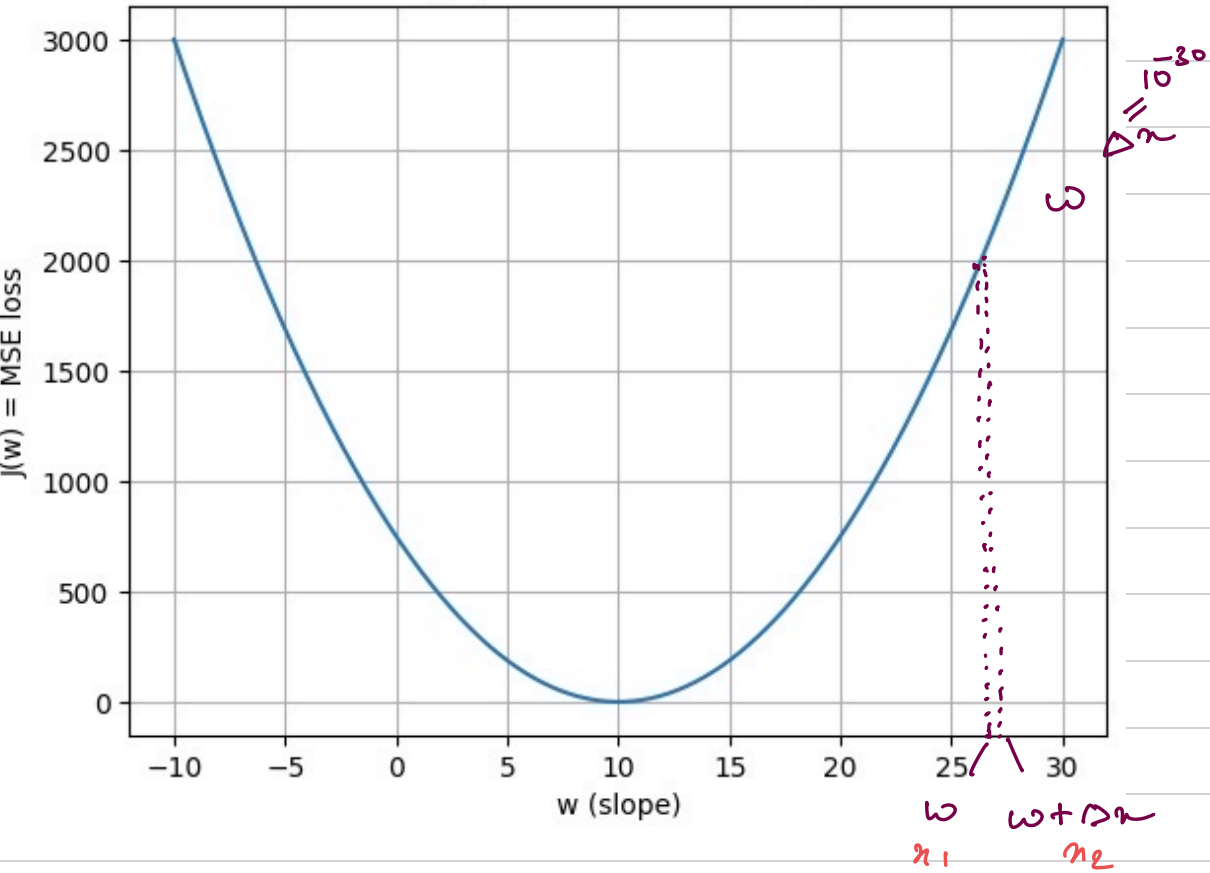




if  $\uparrow w \rightarrow \text{Loss} \downarrow$   
 $\downarrow w \rightarrow \text{Loss} \downarrow$



Loss  $J(w)$  as a function of slope  $w$



$$y_1 \quad y_2$$

$$L(w) ; L(w + \Delta w)$$

$$\frac{L(w + \Delta w) - L(w)}{(w + \Delta w) - w} \Rightarrow \frac{L(w + \Delta w) - L(w)}{\Delta w}$$

$$\Delta w \rightarrow 0$$

$$f(x) = x^2 ; \quad \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\Rightarrow \frac{(x + \Delta x)^2 - x^2}{\Delta x} \Rightarrow \frac{x^2 + \Delta x^2 + 2x\Delta x - x^2}{\Delta x}$$

$$(a+b)^2 = a^2 + b^2 + 2ab \Rightarrow \frac{\Delta x^2 + 2x\Delta x}{\Delta x}$$

$$\Rightarrow \frac{\cancel{\Delta x}(\Delta x + 2x)}{\cancel{\Delta x}}$$

$$= \Delta x + 2x ; \quad \Delta x \rightarrow 0$$

$$= 2x$$



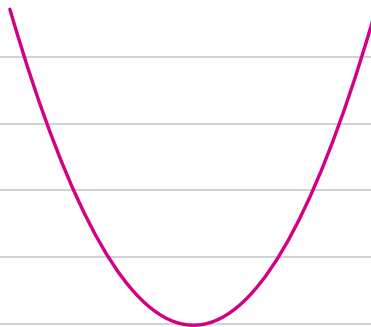
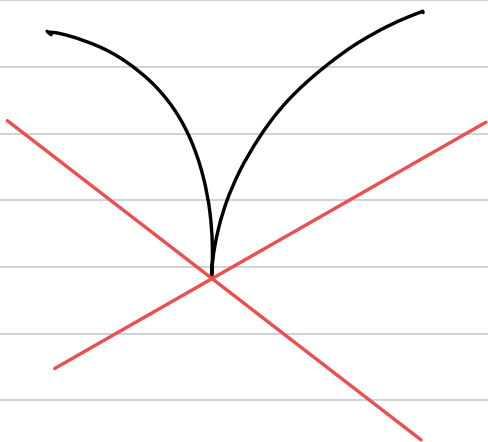
$$f(x+\Delta x) = x^2 + x\Delta x + \Delta x^2 + x\Delta x$$

$$f(m) = m^2$$

$$\frac{2n \, dn + dn^2}{dn}$$

$$\leq \frac{x \cdot dx + dx^2 + x dx}{\Delta x}$$

$$\therefore \lim_{dn} \frac{(2n + dn)}{dn} \rightarrow 2n + dn ; dn \rightarrow 0$$
$$= 2n$$



- $\frac{d}{dx} x^n = nx^{n-1}$
- $\frac{d}{dx} (fg) = fg' + gf'$
- $\frac{d}{dx} \left( \frac{f}{g} \right) = \frac{gf' - fg'}{g^2}$
- $\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$
- $\frac{d}{dx} (\sin x) = \cos x$
- $\frac{d}{dx} (\cos x) = -\sin x$
- $\frac{d}{dx} (\tan x) = \sec^2 x$
- $\frac{d}{dx} (\cot x) = -\csc^2 x$
- $\frac{d}{dx} (\sec x) = \sec x \tan x$
- $\frac{d}{dx} (\csc x) = -\csc x \cot x$
- $\frac{d}{dx} (e^x) = e^x$
- $\frac{d}{dx} (a^x) = a^x \ln a$
- $\frac{d}{dx} \ln x = \frac{1}{x}$
- $\frac{d}{dx} (\arcsin x) = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx} (\arctan x) = \frac{1}{1+x^2}$

$$\textcircled{1} \quad \frac{d(n^3)}{dn} = 3 \cdot n^{3-1} = 3 \cdot n^2$$

$$\textcircled{2} \quad \underbrace{2n^2 + 3n} \quad \underbrace{3n \frac{da}{dn} = 3}$$

$$2 \frac{d(n^2)}{dn} = 2 \cdot n^{2-1} = 2n ; 2n+3$$

$n$

$$\frac{1}{n} \Rightarrow \frac{1}{1} = 1$$

$$3n = 3 \times n^1$$

$$1 \cdot n^{1-1} = 1 \cdot n^0 = 1$$

$\textcircled{3}$

$$\underbrace{3n^2 + 5}$$

$$3 \times 2n^{2-1}$$

$$3 \times 2n$$

$$\frac{d(5)}{dn} = 0$$

$$= 6n + 0$$

$\textcircled{4}$

$$\underbrace{n^3}_1 - \underbrace{2n^2}_{2 \times 2 \times n} + \underbrace{n}_1 - 20$$

$$3n^2 - 4n + 1$$

⑤

$$\frac{d}{dx}(fg) = fg' + gf'$$

$$x \cdot \sin x$$

Diagram showing the product rule components:  $x$  is labeled  $f$  and  $\sin x$  is labeled  $g$ .

$$x \cdot \frac{d \sin x}{dx} + \sin x \cdot \frac{dx}{dx}$$

$$x \cdot \cos x + \sin x$$

